

Instawards Statement of Work (SOW) 30-Day Scoped Engagement

1. Project & Team Information

Project Name:	Byetery
Builder / Team Name:	Byetery
Primary Contact (Name + Email):	Christian Medina - email pending
Ambassador Chapter:	Chile
Ambassador Chapter Lead:	To be confirmed
Suggested Sprint Start Date:	Start within 3-5 business days of approval

2. Instawards Overview & Intent

2.1 Instawards Purpose (for Builder Context)

This engagement defines specific technical outcomes to be completed and demonstrated within 30 days. Byetery will build a battery traceability and incentive MVP on Stellar Testnet, linking unique battery IDs to authorized collection and recycling records.

Detailed operational information will remain off-chain. Soroban will record relevant lifecycle states, actor authorizations and cryptographic commitments to supporting records. A Stellar Testnet asset will demonstrate the reward mechanism.

2.2 Why Stellar

Stellar provides a shared verification layer between independent participants without requiring them to rely solely on Byetery's database. Soroban is used for authorized lifecycle attestations and integrity commitments, while Stellar assets enable programmable rewards after verified recycling events.

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3. Problem Statement & Objective

Problem Being Addressed	What specific gap will this work address?	Battery return and recycling records can be fragmented across consumers, collection points and recyclers, making it difficult to verify what happens to a battery after it reaches the end of its useful life. This limits traceability, reduces accountability between independent participants and makes it difficult to link responsible returns with user incentives. Byetery addresses this gap by creating a shared, independently inspectable history linking an identified battery to authorized collection and recycling attestations.
Objective of This Instaward	What will be true at the end of 30 days?	Deliver a functional web MVP demonstrating at least one complete battery lifecycle on Stellar Testnet: registration, return, authorized collection, authorized recycling confirmation and a test reward, with transaction evidence.

4. Scope of Work (30-Day Deliverables)

The proposed scope is limited to a demonstrable technical MVP within 30 calendar days. The demonstration may use a simulated battery and simulated operational records; it does not establish a production recycling service.

4.1 In-Scope Deliverables

Deliverable	Description (What will be built or produced?)	Why this matters
Deliverable 1 Verifiable battery traceability protocol	Develop and deploy a Soroban contract on Stellar Testnet. Support unique battery registration, lifecycle states, authorized actor roles, collection and recycling confirmations, and hashes of supporting off-chain records. Restrict state changes to the appropriate administrator, collection point or recycler. Reject duplicate IDs and invalid lifecycle transitions. Expose the current state and transaction evidence.	Creates a verifiable record of who attested to a lifecycle event and protects relevant state changes through authorization. Keeps detailed operational data out of the contract.
Deliverable 2 Byetery web MVP and off-chain data layer	Build a web application and database connected to the contract. Register battery ID, type, manufacturer or brand, batch and registration date; generate a QR identifier. Allow QR or ID lookup, lifecycle history and links to Stellar transactions. Provide role-based interfaces for registration, collection and recycling. Compare hashes of versioned supporting records with their on-chain commitments.	Makes the traceability process usable by participants and demonstrates the connection between application records and Stellar evidence.

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Deliverable	Description (What will be built or produced?)	Why this matters
Deliverable 3 Verified recycling and digital reward flow	Execute at least one complete Register - Return - Collect - Recycle - Reward demonstration. Release a Stellar Testnet asset reward after an authorized recycler confirms the recycling event. Associate the reward with the battery and recipient; prevent a second reward for the same completed lifecycle. Record transfer status and transaction evidence.	Demonstrates an incentive linked to an authorized lifecycle attestation. Test assets have no stated monetary value and do not constitute a production token launch.

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4.1 In-Scope Deliverables (continued)

Architecture and verification boundaries

Application and database: QR / battery ID interaction, operational metadata, user accounts, supporting records and lifecycle history. Proposed stack: React / Next.js, Node.js API and PostgreSQL / Supabase, with authentication appropriate to the actor roles.

Soroban on Stellar Testnet: battery identifiers, relevant lifecycle states, authorized actor references and hashes of supporting records. The application will expose the corresponding contract ID and transaction hashes for review.

Reward: a Stellar Testnet asset sent to a test wallet once the recycling confirmation condition is met. Reward completion is tracked separately from the physical lifecycle state so a failed transfer does not imply an unpaid user has been rewarded.

For the MVP, the lifecycle is Registered - Returned - Collected - Recycled. A return request alone does not authorize a reward. The collection point confirms receipt, and the recycler confirms recycling. Distribution tracking is outside the required demonstration.

A matching hash proves that a disclosed record matches its anchored version; it does not independently prove that physical recycling occurred. The physical claim depends on the authorized actor and its supporting evidence. Simulated evidence will be labeled clearly. Personal data will remain off-chain.

Out-of-Scope (Explicitly Not Included)

Stellar Mainnet deployment; production GREEN token economics, public token sales or exchange listings; production USDC payments or custody infrastructure; native mobile apps; IoT, NFC or RFID hardware; automated recycling machinery; logistics optimization; industrial-scale or nationwide deployment; enterprise analytics; production SLAs; carbon credits; battery NFTs; legal or environmental certification; and commercial agreements with manufacturers or recyclers.

4.2 Deliverable-Aligned Budget Request

Requested Budget Amount	Rationale for Budget Request
\$4,900 USD Proposed budget	Technical implementation labor across the three deliverables: Soroban contract and Stellar integration \$1,700; backend, database and API \$1,100; frontend and QR interface \$900; reward integration and end-to-end testing \$700; documentation, deployment and demo \$500. Total: \$4,900. Contributor allocation remains to be finalized. No production reward pool is included.

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5. 30-Day Execution Plan & Timeline

5.1 Weekly Breakdown

Week	Planned Work	Expected Output
Week 1 Days 1-7	Define lifecycle model, record hashing and actor permissions. Implement and test registration, collection, recycling and authorization in Soroban.	Testnet contract deployed; valid lifecycle transactions and rejected unauthorized or invalid transitions.
Week 2 Days 8-14	Build database, API and web interfaces. Connect contract calls, QR generation, battery lookup and history.	Working web MVP with battery records and corresponding Stellar verification.
Week 3 Days 15-21	Integrate authorized collection and recycling workflows, test wallet and reward transfer. Test duplicate reward prevention and failure handling.	Complete lifecycle demonstration, including a successful reward and verifiable transaction hashes.
Week 4 Days 22-30	Fix defects, polish the interface, deploy the demonstration and publish repository documentation, architecture and demo recording.	Accessible MVP, reviewable repository, demo video and completion evidence.

6. Evidence of Completion (Required)

Evidence will be organized by deliverable and accompanied by concise review instructions so the Ambassador Chapter Lead can verify the result without extensive technical knowledge.

6.1 Planned Evidence to Be Submitted

Deliverable	Evidence Type	Description
Deliverable 1	GitHub repository, contract ID, Testnet transaction hashes, test results and contract documentation.	Shows registration, authorized lifecycle changes, metadata commitments and rejection of invalid operations.
Deliverable 2	MVP URL, demo video, screenshots, QR example and repository.	Shows battery creation, lookup, lifecycle history and verification against Stellar records.
Deliverable 3	End-to-end video, lifecycle record, reward transaction hash and test results.	Shows the complete lifecycle, recipient transfer and prevention of a duplicate reward.

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6.2 Evidence Verification Checklist (For Ambassador Use)

For each deliverable, the Ambassador Chapter Lead will assess whether the evidence is present and sufficient.

Deliverable	Evidence Present	Evidence Partial	Evidence Missing	Comments
Deliverable 1	[]	[]	[]	
Deliverable 2	[]	[]	[]	
Deliverable 3	[]	[]	[]	

Demonstration scenario and acceptance criteria

Register battery **BYE-000001**, type lithium-ion, batch TEST-001, and generate its QR code. Associate it with a simulated user and test wallet. Record the return; Collection Point 01 confirms receipt, and Recycler 01 submits a labeled test recycling record. Anchor the record hash through Soroban.

Following the authorized recycling confirmation, transfer an illustrative **10 GREEN-TEST** units to the test wallet. This is a demonstration amount, not a commitment to future token economics. Display Registered, Returned, Collected, Recycled and reward payment status with transaction links.

Completion requires a unique ID, role-restricted state changes, a readable history, matching supporting-record hashes and a successful reward transaction. Negative tests must show that an unauthorized actor cannot confirm recycling, an altered record fails hash verification, and a repeated request cannot issue a second reward for the same lifecycle.

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7. Next-Step Alignment

7.1 Anticipated Next Step After Completion

After successful completion, Byetery intends to pursue a small real-world pilot with one or two collection points and a recycling partner. Partner participation, operating procedures and future incentives will be evaluated before handling real battery flows.

Funding and development options to confirm with the Ambassador Chapter Lead:

- ☐ Apply to SCF Build Award
- ☐ Continue development independently
- ☐ Apply for a follow-on Instaward (if eligible)
- ☐ Seek other ecosystem support
- ☐ Other: local pilot with collection and recycling partners

8. Instawards Constraints Acknowledgement

The following acknowledgement fields are retained from the supplied reference SOW for review before submission. They are left unchecked for the Builder to confirm. Program limits below are reproduced from that reference and have not been independently verified as current.

- ☐ This scope will be completed within 30 days or less.
- ☐ Instawards support execution, not open-ended exploration.
- ☐ A project may receive no more than two follow-on Instawards.
- ☐ Each Instaward is capped at \$5,000.
- ☐ Total Instawards funding may not exceed \$15,000.

9. Submission Confirmation

Once finalized, this Statement of Work is intended to be submitted by the Ambassador Chapter Lead via the Instawards Airtable submission form for review and approval, following the process described in the supplied reference.

Before submission, confirm the primary contact email, Ambassador Chapter Lead, final budget allocation and applicable program requirements. This document defines proposed future work; it does not claim completed development, secured recycling partnerships, funding approval or an already submitted application.